

Removing the finite-dimensional Hilbert restriction in a polynomial $\mathbf{L}_{p,p}$ theorem

FA/Banach run packet for arXiv:2209.01767

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Source signal

The source paper is S. Dantas, M. Jung, M. Mazzitelli, and J. T. Rodriguez, *On the strong subdifferentiability of the homogeneous polynomials and (symmetric) tensor products*, arXiv:2209.01767.

In Theorem 4.7 the authors prove that, in the complex case, the pair (c_0, H) has the 2-homogeneous polynomial $\mathbf{L}_{p,p}$ whenever H is a finite-dimensional Hilbert space. Immediately after the proof they observe that finite dimensionality of H was used only to know that (ℓ_∞^A, H) has the 2-homogeneous polynomial $\mathbf{L}_{p,p}$, and that if this remained true for infinite-dimensional Hilbert spaces then Theorem 4.7 would also hold for them.

simply note that if X, Y are finite dimensional Banach spaces, then the pair (X, Y) has the 2-homogeneous polynomial $\mathbf{L}_{p,p}$. Indeed, it follows by contradiction using the compactness of the unit ball $B_{\mathcal{P}(\ell_\infty^A(X,Y))}$.

Theorem 4.7. For a finite dimensional Hilbert space H , the pair (c_0, H) has the 2-homogeneous polynomial $\mathbf{L}_{p,p}$ in the complex case.

Proof. Let $\varepsilon > 0$ and $x_0 \in S_{c_0}$ be fixed. Consider the finite set $A := \{i \in \mathbb{N} : |x_0(i)| = 1\}$. Since ℓ_∞^A and H are finite dimensional, we can consider $\tilde{\eta}(\varepsilon, x_0^A) > 0$ from the 2-homogeneous polynomial $\mathbf{L}_{p,p}$ for the pair (ℓ_∞^A, H) . We may assume that $(1 - \varepsilon)\tilde{\eta}(\varepsilon, x_0^A) < \varepsilon$. Suppose that

$$\|P(x_0)\| > 1 - \min \left\{ \eta \left(\min \left\{ \frac{(1 - \varepsilon)\tilde{\eta}(\varepsilon, x_0^A)}{2}, \varepsilon \right\}, x_0 \right), \frac{(1 - \varepsilon)\tilde{\eta}(\varepsilon, x_0^A)}{2} \right\},$$

where $\eta(\cdot, x_0) > 0$ is chosen from Proposition 4.6. Then we have that

$$\|P - P \circ P_A\| < \min \left\{ \frac{(1 - \varepsilon)\tilde{\eta}(\varepsilon, x_0^A)}{2}, \varepsilon \right\}.$$

Thus,

$$\begin{aligned} \|P \circ P_A(x_0)\| &= \|P(x_0) + (P_A(x_0) - P(x_0))\| \\ &> \left(1 - \frac{(1 - \varepsilon)\tilde{\eta}(\varepsilon, x_0^A)}{2} \right) - \|P - P_A\| \\ &> \left(1 - \frac{(1 - \varepsilon)\tilde{\eta}(\varepsilon, x_0^A)}{2} \right) - \frac{(1 - \varepsilon)\tilde{\eta}(\varepsilon, x_0^A)}{2} = 1 - (1 - \varepsilon)\tilde{\eta}(\varepsilon, x_0^A). \end{aligned}$$

This implies that there exists $\bar{Q} \in \mathcal{P}(\ell_\infty^A, H)$ with $\|\bar{Q}\| = 1$ such that

$$\|\bar{Q}(x_0^A)\| = 1 \quad \text{and} \quad \|\bar{Q} - P \circ P_A\| < \varepsilon,$$

where $P \circ P_A$ is viewed as an element of $\mathcal{P}(\ell_\infty^A, H)$. Define $Q \in \mathcal{P}(c_0, H)$ as the natural extension of $\bar{Q}$ to c_0 , that is, $Q(x) = \bar{Q}(x^A)$. Note that $\|Q\| = \|Q(x_0)\| = 1$ and

$$\|Q - P\| \leq \|\bar{Q} - P \circ P_A\| + \|P \circ P_A - P\| < 2\varepsilon,$$

which completes the proof. $\square$

Notice that the hypothesis of H being finite dimensional was only used to affirm that the pair (ℓ_∞^A, H) has the 2-homogeneous polynomial $\mathbf{L}_{p,p}$. If this held for infinite dimensional Hilbert spaces, so it would do Theorem 4.7.

Definition

We use the source paper's local Bishop–Phelps–Bollobas point property for homogeneous polynomials. A pair (X, Y) has the N -homogeneous polynomial $\mathbf{L}_{p,p}$ if for every $\varepsilon > 0$ and $x_0 \in S_X$ there is $\eta(\varepsilon, x_0) > 0$ such that, whenever $P \in S_{\mathcal{P}(^N X, Y)}$ and $\|P(x_0)\| > 1 - \eta(\varepsilon, x_0)$, there is $Q \in S_{\mathcal{P}(^N X, Y)}$ with

$$\|Q(x_0)\| = 1 \quad \text{and} \quad \|Q - P\| < \varepsilon.$$

This is the norm-one version used in the proof of Theorem 4.7.

Idea of the proof

The missing infinite-dimensional Hilbert case is actually finite-dimensional after one looks at a single polynomial. If E is finite dimensional, then a polynomial $P : E \rightarrow H$ is determined by finitely many Hilbert-valued coefficients. Therefore the linear span of $P(E)$ has dimension bounded only by N and $\dim E$. Hilbert spaces have only one isometry type in each finite dimension, so the compactness proof used for finite-dimensional range spaces is uniform over all such coefficient spans. This proves (E, H) has the polynomial $\mathbf{L}_{p,p}$ for every Hilbert space H .

Applying this with $E = \ell_\infty^A$, where A is the finite set of coordinates at which $x_0 \in S_{c_0}$ has modulus one, supplies exactly the ingredient the source proof of Theorem 4.7 said was missing.

Finite-dimensional domains

Lemma 1. *Let E be a finite-dimensional Banach space, let H be a Hilbert space over the same scalar field, and let $N \in \mathbb{N}$. Then the pair (E, H) has the N -homogeneous polynomial $\mathbf{L}_{p,p}$.*

Proof. Fix $\varepsilon > 0$ and $x_0 \in S_E$. Let

$$M = \dim \mathcal{P}({}^N E, \mathbb{K}).$$

Every $P \in \mathcal{P}({}^N E, H)$ has range contained in a Hilbert subspace of dimension at most M : choose a basis $\phi_1, \dots, \phi_M$ of $\mathcal{P}({}^N E, \mathbb{K})$ and write

$$P(x) = \sum_{j=1}^M \phi_j(x) h_j$$

for suitable $h_j \in H$.

We first note that for each $1 \leq k \leq M$, the pair (E, ℓ_2^k) has the N -homogeneous polynomial $\mathbf{L}_{p,p}$. Indeed, if not, there are $P_m \in S_{\mathcal{P}({}^N E, \ell_2^k)}$ with $\|P_m(x_0)\| \rightarrow 1$ and no norm-one polynomial within ε of P_m attaining norm at x_0 . Since $\mathcal{P}({}^N E, \ell_2^k)$ is finite dimensional, after passing to a subsequence $P_m \rightarrow P$ in norm. Then $\|P\| = 1$ and $\|P(x_0)\| = 1$, so P itself is a norm-one polynomial attaining at x_0 , contradicting $\|P_m - P\| \rightarrow 0$.

Let $\eta_k > 0$ be a corresponding constant for (E, ℓ_2^k) , and put $\eta = \min_{1 \leq k \leq M} \eta_k$. Now take $P \in S_{\mathcal{P}({}^N E, H)}$ with $\|P(x_0)\| > 1 - \eta$. Let $F = \text{span } P(E)$, let $k = \dim F$, and choose a linear isometry $U : F \rightarrow \ell_2^k$. Then UP belongs to $S_{\mathcal{P}({}^N E, \ell_2^k)}$ and $\|(UP)(x_0)\| > 1 - \eta_k$. Hence there is $R \in S_{\mathcal{P}({}^N E, \ell_2^k)}$ such that $\|R(x_0)\| = 1$ and $\|R - UP\| < \varepsilon$. The polynomial $Q = U^{-1}R : E \rightarrow F \subset H$ satisfies

$$\|Q\| = 1, \quad \|Q(x_0)\| = 1, \quad \|Q - P\| < \varepsilon.$$

Thus (E, H) has the desired property. □

Main theorem

Theorem 1. *Let H be an arbitrary complex Hilbert space. Then the pair (c_0, H) has the 2-homogeneous polynomial $\mathbf{L}_{p,p}$.*

Proof. We follow the proof of Theorem 4.7 in the source paper, replacing its finite-dimensional Hilbert input by Lemma 1.

Fix $x_0 \in S_{c_0}$ and $\varepsilon > 0$. Let

$$A = \{i \in \mathbb{N} : |x_0(i)| = 1\}.$$

This set is finite because $x_0 \in c_0$. Proposition 4.6 of the source paper applies to every complex Hilbert space H : for the chosen x_0 and each $\delta > 0$, if $P \in S_{\mathcal{P}(2c_0, H)}$ nearly attains its norm at x_0 , then

$$\|P - P \circ P_A\| < \delta,$$

where $P_A : c_0 \rightarrow \ell_\infty^A$ is the coordinate projection.

By Lemma 1, the pair (ℓ_∞^A, H) has the 2-homogeneous polynomial $\mathbf{L}_{p,p}$. Choose the same constants as in the source proof: first take the $\mathbf{L}_{p,p}$ modulus $\tilde{\eta}(\varepsilon, x_0^A)$ for (ℓ_∞^A, H) , then use Proposition 4.6 with

$$\delta = \min \left\{ \frac{(1 - \varepsilon)\tilde{\eta}(\varepsilon, x_0^A)}{2}, \varepsilon \right\}.$$

If $P \in S_{\mathcal{P}(2c_0, H)}$ is sufficiently close to attaining at x_0 , then $\|P - P \circ P_A\| < \delta$, and hence

$$\|(P \circ P_A)(x_0)\| > 1 - (1 - \varepsilon)\tilde{\eta}(\varepsilon, x_0^A).$$

The $\mathbf{L}_{p,p}$ property of (ℓ_∞^A, H) gives a norm-one polynomial $\overline{Q} \in \mathcal{P}(2\ell_\infty^A, H)$ such that

$$\|\overline{Q}(x_0^A)\| = 1 \quad \text{and} \quad \|\overline{Q} - P \circ P_A\| < \varepsilon.$$

Extend it to c_0 by $Q(x) = \overline{Q}(P_A x)$. Then

$$\|Q\| = \|Q(x_0)\| = 1$$

and

$$\|Q - P\| \leq \|\overline{Q} - P \circ P_A\| + \|P \circ P_A - P\| < 2\varepsilon.$$

Renaming 2ε as the requested tolerance proves the 2-homogeneous polynomial $\mathbf{L}_{p,p}$ for (c_0, H) . $\square$

Corollary 1. *Let H be an arbitrary complex Hilbert space. Then*

$$(\widehat{\otimes}_{\pi_s}^2 c_0) \widehat{\otimes}_\pi H$$

is strongly subdifferentiable on the set

$$V = \{\otimes^2 x \otimes y : \|x\| = \|y\| = 1\}.$$

Proof. Hilbert spaces have micro-transitive norm, and H^* is again a Hilbert space. The source paper proves that, when Y has micro-transitive norm, (X, Y^*) has the N -homogeneous polynomial $\mathbf{L}_{p,p}$ if and only if $(\widehat{\otimes}_{\pi_s}^N X) \widehat{\otimes}_\pi Y$ is SSD on $\{\otimes^N x \otimes y : \|x\| = \|y\| = 1\}$. Apply this with $X = c_0$, $Y = H$, $N = 2$, and Theorem 1. $\square$

Verification and limitations

The only new point is Lemma 1. Its proof uses two elementary facts: a homogeneous polynomial on a finite-dimensional domain has range contained in a uniformly bounded finite-dimensional coefficient span, and all k -dimensional Hilbert spaces over the same scalar field are linearly isometric. Therefore the finite-dimensional compactness argument used by the source authors applies uniformly after passing through the coefficient span.

This packet does not address the separate question in Remark 2.7(iii) of the source paper about replacing uniform convexity of Y in Theorem A by the assumption that Y^* is SSD. It also does not address the broad open problem of whether there are non-Hilbert Banach spaces with micro-transitive norms. Its scope is the Hilbert-codomain restriction in Theorem 4.7 and the corresponding finite-dimensional restriction in the following tensor-product corollary.

Novelty check

The cheap run indexes were searched for arXiv:2209.01767, strong subdifferentiability, homogeneous polynomials, $L_{p,p}$, c_0 , Hilbert codomain, and symmetric tensor products. No existing packet for this paper or this finite-dimensional Hilbert restriction was found. Adjacent run entries concern other homogeneous-polynomial questions and isometric homogeneity, not this source theorem. External novelty confidence is bounded by the local source/index search performed here.

References

- S. Dantas, M. Jung, M. Mazzitelli, and J. T. Rodriguez, *On the strong subdifferentiability of the homogeneous polynomials and (symmetric) tensor products*, arXiv:2209.01767.

Human-review recommendation

Review as a likely valid partial result. The main point to check is that an H -valued polynomial on a finite-dimensional domain has range contained in a finite-dimensional Hilbert subspace of dimension bounded by $\dim \mathcal{P}({}^N E, \mathbb{K})$, so the source paper's finite-dimensional compactness constant can be chosen after reducing to one of finitely many Hilbert dimensions.